

Xinyun (Mark) Zhou

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

BS in Computer Science, Manning College of Information and Computer Sciences (CICS)

Sep. 2026 – May 2030

- **Chancellor's Award** (merit-based admission scholarship)

George School

Newtown, PA

High School Diploma, College/University Preparatory Program

Sep. 2021 – May 2026

- Took gap year 2024-2025 to focus full-time on competitive programming, machine learning, robotics, and heuristic optimization, winning multiple university contests
- **Relevant Coursework:** Advanced Programming: Artificial Intelligence, Statistics with Data Science, Proof & Advanced Topics in Mathematics, Advanced Math: Linear Algebra, AP Calculus BC, Advanced Physical Computing & Robotics

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Swift/SwiftUI, HTML/CSS/JavaScript

Frameworks & Tools: PyTorch, Git, Xcode, iOS Development, CAD, Teensy/Embedded Systems

PROJECTS

Tiaoqi Master – Chinese Checkers iOS App

github.com/markjough/tiaoqi-master

Swift Student Challenge 2026 Winner

2025 – 2026

- Designed and built a native iOS app in Swift 6 and SwiftUI that teaches Chinese Checkers through a play-as-you-learn tutorial and an adaptive AI opponent that scales to the player's skill level
- Implemented an iterative-deepening minimax engine with alpha-beta pruning, heuristic move ordering, and a time-budgeted search over an axial hex-coordinate board for responsive on-device play
- Authored a contextual commentary system that reacts to chain length, forward progress, and setup play, backed by a procedural audio-synthesis engine built on AVFoundation

RotateMate – Image Rotation ML Model

github.com/markjough/RotateMate-model

Personal Project

2025 – Present

- Built an end-to-end iOS ML system to automatically detect and correct mis-rotated images, spanning custom dataset creation, PyTorch training, and Swift/SwiftUI deployment
- Optimized for on-device efficiency: designed 3M parameter model, implemented quantization pipeline, and engineered hardware-accelerated preprocessing to achieve <1ms inference latency on modern NPUs

Competitive Programming Library

github.com/markjough/cp-library

Open Source Project

2023 – Present

- Authored a C++ library of advanced data structure and algorithm implementations for Codeforces, ICPC, and educational reference

RESEARCH

Research Assistant, Gerych Lab

Worcester Polytechnic Institute

Trustworthy ML (Remote); Advisor: Prof. Walter Gerych

Apr. 2026 – Present

- Contributing to research on test-time debiasing of CLIP-style vision-language models, building on the lab's BendVLM (NeurIPS 2024) and WRING work on concept-subspace bias amplification
- Onboarding via advisor-curated reading on CLIP/VLM internals and concept-subspace debiasing; preparing evaluation work on projection- vs rotation-based methods
- Expected middle-author contribution on a forthcoming paper

EXPERIENCE

Algorithms Club – President & Co-Founder

George School

Leadership Position

Sep. 2024 – Present

- Co-founded and lead school's Algorithms Club with 30+ active members
- Organize weekly meetings covering competitive programming techniques, algorithm design, and problem-solving strategies
- Helped members advance through USACO divisions, up to Platinum (the contest's highest)

Math Tutor

George School

Volunteer Position

Sep. 2024 – Present

- Provide weekly peer tutoring in advanced mathematics including combinatorics and other topics
- Help students develop problem-solving skills and mathematical intuition through one-on-one instruction

HONORS & AWARDS

Swift Student Challenge Winner

Apple

Selected as a 2026 Swift Student Challenge winner for Tiaoqi Master, a SwiftUI app teaching Chinese Checkers 2026

USACO Platinum

USA Computing Olympiad

Highest division of premier US high school algorithmic competition

Mar. 2025

Codeforces Master (Rating: 2219)

Codeforces

Top 1% of competitors worldwide in timed algorithmic contests

Aug. 2024

First Place – CMIMC Programming Contest

Carnegie Mellon University

1st overall and 1st in Optimization round, team-based algorithmic challenges

Apr. 2025

First Place – Massachusetts Computer Science Olympiad

MACSO

Won in-person high school finals with 278 participants from 182 schools in 40 countries

Nov. 2024

First Place – PClassic Programming Contest (Advanced)

University of Pennsylvania

1st overall in Advanced Division, four-hour algorithm and data structure contest

Apr. 2024

First Place – TJ Invitational Open in Informatics (TJIOI)

Thomas Jefferson High School

1st overall in two-person team, multi-hour algorithmic coding challenges

Apr. 2024